

BASH SCRIPTING LOOPS BASICS

Bash for loop

```
1  for i in /etc/*; do
2      echo $i
3  done
4
5  # Same as above(alternate
6  syntax), also works with other
7  loop structs
8
9  for i in /etc/*
10 do
11     echo $i
12 done
```

C-like for loop

```
1  for ((i = 0 ; i < 100 ; i++)); do
2      echo $i
3  done
4
5  # Same as above (alternate
6  syntax) also works with other
7  loop structs
8
9  for ((i = 0 ; i < 100 ; i++))
10 do
11     echo $i
12 done
```

For loop ranges

```
1  for i in {1..10}; do
2      echo "Number: $i"
3  done
4
5  # With step size
6  # => {START..STOP..STEP}
7
9  for i in {5..50..5}; do
10
11     echo "Number: $i"
12
13 done
```

Bash while loop

```
1  # incrementing the value
2  i=1
3  while [[ $i -lt 4 ]]; do
4      echo "Number: $i"
5      ((i++))
6  done
7
8  # decrementing the value
9
10 i=3
11 while [[ $i -gt 0 ]]; do
12     echo "Number: $i"
13     ((i--))
14 done
```

Bash while True loop

```
1  # while true long hand
2
3  while true; do
4      # TODO
5      # TODO
6  done
7
8  # or the shorthand (alternate
9  syntax)
10
11 while ;; do
12     # TODO
13     # TODO
14 done
```

Reading files

```
1  # using pipes
2
3  cat file.txt | while read line
4  do
5      echo $line
6  done
7
8
9  # OR using input redirection
10
11
12 while read line; do
13     echo $line
14 done < "/path/to/txt/file"
```

Continue statement

```
1  # seq command can be used to
2  generate ranges
3
4  for number in $(seq 1 3); do
5
6      if [[ $number = 2 ]];
7      then
8          continue;
9      fi
10
11     echo "$number"
12
13 done
```

Break statement

```
1  for number in $(seq 1 3); do
2
3      if [[ $number = 2 ]]; then
4
5          # Skip entire rest of
5          loop or break out
7          of the loop.
8
9          break;
10
11     fi
11     # This will only print 1
12     echo "$number"
13 done
```

Until or do loop

```
1  # incrementing
2  count=0
3  until [ $count -gt 10 ]; do
4      echo "$count"
5      ((count++))
6  done
7
8  # decrementing
9  count=10
10 until [ $count -eq 0 ]; do
11     echo "$count"
12     ((count--))
13 done
```